GLUEING OBJECTS OF THE DERIVED CATEGORY ON A TWO-OPEN COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Mayer-Vietoris for Hom. Cohomology of Sheaves, Lemma cohomology-lemma-mayer-vietoris-hom. Long exact sequence for Hom on a two-open covering.
- Compatibility of restriction and pushforward. Cohomology of Sheaves, Section cohomology-section-locality. Canonical identifications (Rj_{V,*}B)|_U =~ Rj_*(B|_{U cap V}).

Source reference. Original lemma 08BW: sources/stacks-chow/tags.txt, source lines 10581–10581. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $X = U \cup V$ be the union of two open subspaces of X. Suppose given*

(1) *an object A of $D(\mathcal{O}_U)$,*
(2) *an object B of $D(\mathcal{O}_V)$, and*
(3) *an isomorphism $c : A|_{U\cap V} \to B|_{U\cap V}$.*

Then there exists an object F of $D(\mathcal{O}_X)$ and isomorphisms $f : F|_U \to A$, $g : F|_V \to B$ such that $c = g|_{U\cap V} \circ f^{-1}|_{U\cap V}$. Moreover, given

(1) *an object E of $D(\mathcal{O}_X)$,*
(2) *a morphism $a : A \to E|_U$ of $D(\mathcal{O}_U)$,*
(3) *a morphism $b : B \to E|_V$ of $D(\mathcal{O}_V)$,*

such that

$$a|_{U\cap V} = b|_{U\cap V} \circ c.$$

Then there exists a morphism $F \to E$ in $D(\mathcal{O}_X)$ whose restriction to U is $a \circ f$ and whose restriction to V is $b \circ g$.

Proof. Denote j_U, j_V, $j_{U\cap V}$ the corresponding open immersions. Choose a distinguished triangle

$$F \to Rj_{U,*}A \oplus Rj_{V,*}B \to Rj_{U\cap V,*}(B|_{U\cap V}) \to F[1]$$

where the map $Rj_{V,*}B \to Rj_{U\cap V,*}(B|_{U\cap V})$ is the obvious one and where $Rj_{U,*}A \to Rj_{U\cap V,*}(B|_{U\cap V})$ is the composition of $Rj_{U,*}A \to Rj_{U\cap V,*}(A|_{U\cap V})$ with $Rj_{U\cap V,*}c$. Restricting to U we obtain

$$F|_U \to A \oplus (Rj_{V,*}B)|_U \to (Rj_{U\cap V,*}(B|_{U\cap V}))|_U \to F|_U[1]$$

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Denote $j : U \cap V \to U$. Compatibility of restriction to opens and cohomology shows that both $(Rj_{V,*}B)|_U$ and $(Rj_{U\cap V,*}(B|_{U\cap V}))|_U$ are canonically isomorphic to $Rj_*(B|_{U\cap V})$. Hence the second arrow of the last displayed diagram has a section, and we conclude that the morphism $F|_U \to A$ is an isomorphism. Similarly, the morphism $F|_V \to B$ is an isomorphism. The existence of the morphism $F \to E$ follows from the Mayer-Vietoris sequence for Hom, see Lemma 08BW. $\square$